

NOAIS-Form 2 Numerical Subtest

Instructions:

The test contains one main type of item:

Sequence: The numbers follow one or more underlying mathematical operations (such as addition, subtraction, multiplication, division, or exponentiation) and/or logical rules or patterns. Your task is to identify these rules and determine the number(s) that should replace the question mark(s).

Some items may involve more than one rule operating simultaneously. Use all of the information provided in the item to determine the correct answer.

The 25 items:

- 1) 0, 11, 222, 3333, ?
- 2) BEGINS 11, 22, 110, 770, 6930, 97020, ?
- 3) 2, 32, 235, 7532, 235711, ?
- 4) 25, 37, 611, 1117, 1825, 2735, ?
- 5) 11, 21, 23, 83, 815, 4815, ?, 384105
- 6) 111, 248, 3927, 41664, ?
- 7) 79, 194, 1882, 17642, ?, 1708442, ...
- 8) 35, 335, 6210, 181050, 724200, ?
- 9) 17, 102, 204, 816, 5712, 34272, ?
- 10) 73, 37, 47, 114, 456, 228, ?, 5679, ...
- 11) 1, 26, 241, 2072, 05040, 403203, ?
- 12) 1729, 171, 63, ?, 45, 45, ?, ...
- 13) 1, 12, 139, 1, 41, 664, 1, 52, 512, 5, ?, ?
- 14) 013, 169, 482, 539, 454, ?
- 15) $10101=270$, $10110=402$, $11001=500$, $11110=6170$, $100101=876$, ?
- 16) 26, 52, 165, 606, 3003, 18810, ?
- 17) 19, 95, 665, $11305/3$, $22610/3$, ?
- 18) 10, 33, 1300, 6363, 34430, 173493, ?
- 19) 11, 22, 132, 2376, 3920400, ?
- 20) 20, 2215, 151443, 33171179, 133171179, ?
- 21) 32, 6, 44, 633, 1594, 92492, 124974, ?
- 22) 171, 173, 436, 6910, 2115, 6121, ?, ?
- 23) 61, 5357, 714322, 13234376, 6789911471, ?
- 24) ENDS 25.3, ?, 196.9, 20.6, 56.8, 16.5, 95.4, 266.8, ?, 361.3
- 25) $25=90$, $48=6223$, $141=4$, $560=$, ?, $16764=0$, $117341=2$, $938720=$, ...